

Hoang Vu

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EDUCATION

York University GPA 7.71/9 <i>Honours Bachelor of Science in Computer Science</i>	Sep 2024 – Apr 2026 Toronto, ON
Sheridan College GPA 3.78/4 <i>Advanced Diploma in Software Development and Network Engineering</i>	Sep 2020 – Dec 2023 Oakville, ON

PROJECTS

- MindForge** | *FastAPI, React, TypeScript, Supabase, OpenAI, Groq, Hugging Face, Ollama Cloud*
- Built a multi-provider data orchestration pipeline integrating OpenAI, Groq, and Hugging Face APIs with retry logic and fallback routing, **reducing content generation costs by 80%**
 - Automated ingestion and transformation of user learning data to personalize lesson and quiz generation within 3 minutes
- Academic Scheduler** | *FastAPI, React, TypeScript, Supabase, OpenAI, PaddlePaddle, LightGBM*
- Built an end-to-end data pipeline ingesting unstructured syllabus PDFs via PaddleOCR, transforming and loading structured schedule data into Supabase PostgreSQL
 - Engineered features and trained a LightGBM model to predict student workload distribution, **reducing deadline clustering by 40% across 50 student schedules**

EXPERIENCE

- GenAI Engineer** Jun 2026 – Present
Manulife (Tata Consultancy Services) Mississauga, ON
- Architected and scaled end-to-end AI application packaging automation, cutting lifecycle processing time from **13.5 hours to 6 hours per package (55% reduction)** and saving **367+ monthly engineering hours**
 - Designed agentic workflows using **Copilot Studio, Window Installer API, FastAPI, Power Automate, Dataverse** and **Sharepoint** to automate **ServiceNow** request intake, validate documentation, generate MST transform files, packaging scripts and Intunewin file for **Intune** upload, handling different package complexities
- Software Developer** May 2023 – Aug 2023
SIRT Centre Toronto, ON
- Designed and implemented multiplayer menu system in C++ featuring character selection, persistent profile management with real-time session state tracking
 - Diagnosed and resolved race conditions through thread-safety analysis and unit testing, improving system reliability for **10+ concurrent users**
- Software Developer** May 2022 – Dec 2022
Cloud DX Oakville, ON
- Built voice controlled interaction system for HoloLens2, showcased at **ANA Avatar XPRIZE 2022** with live demonstrations
 - Debugged data transmission issue between Arduino sensors and HoloLens2, tracing the root cause through C++/C# communication layer and restoring accurate real-time display
 - Implemented gesture-based interaction for HoloLens2, improving task completion speed by 20% through usability testing
- Software Developer** Jan 2022 – Apr 2022
Medtech Canada Oakville, ON
- Migrated deprecated architecture resulting in **40% faster scene load times**
 - Rapidly resolved a critical rendering bug within **48 hours**, restoring accurate motion-tracking data accuracy

TECHNICAL SKILLS

Languages: Java, C#, Python, C++, TypeScript
Frontend: React, Bootstrap, jQuery
Backend: FastAPI, Spring Boot, ASP.NET, Node, Express
AI/ML: pandas, NumPy, scikit-learn
Database: PostgreSQL, MongoDB, Supabase
Tools: Git, VS Code, Visual Studio, Eclipse